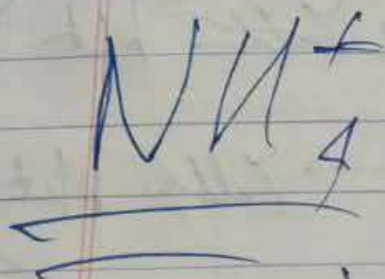


DATE
NO

Test of Basic Radical

Group $\Rightarrow$	0 - NH_4^+ ✓ (dil. HCl) (Resist)
	1 - Pb^{2+} (dil. HCl) (Resist Group 1) ✓
	2 - Cu^{2+} ✓ (dil. HCl + H_2S)
Fe, Ni, Co, Mg	3 - Al^{3+} ✓, Fe^{3+} - X
<u>Coloured.</u>	4 - Zn^{2+} ✓, Mg^{2+} , Ni^{2+} , Co^{2+} - X
	5
	6

0th Group (NH_4^+) $\rightarrow$ Salt + NaOH $\xrightarrow{\Delta}$
 Colorless Gas w/ pungent
 ammonia odour.



+
 Glass Rod in HCl put
 on mouth of $\uparrow$
 which gives dense
 white fumes.

DATE _____
NO _____

Qual Pb²⁺

1. Salt solⁿ + dil HCl → White ppt
Conf. G.I.
Pb²⁺

2. Dissolve ppt in Hot Water & divide into
3 parts.

i) Part 1 :- Let it cool, it shows
Crystals.

ii) Part 2 :- Add KI → Yellow ppt

iii) Part 3 :- Add K₂CrO₄ → Yellow ppt

Confirm for
Pb²⁺

DATE
NO

Group III (M³⁺)
GP Reagent - NH_4Cl (s) + NaOH
(in excess)

DS. $\text{NH}_4\text{Cl} \rightarrow$ Gelatin ppt
white
Blue color test NaOH^+

$\therefore$ Dissolve ppt in dil HCl + Blue litmus
Salt +

* NaOH drop wise
 $\rightarrow$ Blue colored floccy
mass

Grp II (Cu^{2+})

Grp. Reagent - dil $\text{HCl} + \text{H}_2\text{S}$

1. O.S + dil $\text{HCl} \rightarrow$ Black
+ $\text{H}_2\text{S}(\text{g})$ ppt

2. Dissolve the ppt in 50% HNO_3
& divide into 2 parts

i) Part I + $\text{NH}_4\text{OH} \rightarrow$ Blue solⁿ

ii) Part II + $\text{K}_4[\text{Fe}(\text{CN})_6] \rightarrow$ ppt
Charcoal Brown

Cu^{2+} Confirmed

For HS^-

Take Na_2S + dil $\text{H}_2\text{SO}_4 \rightarrow$
 $\text{Gas HS}(\text{g})$

Grp III Fe^{3+}

1. O.S + NH_4Cl + NH_4OH

(1) Dissolve the ppt in dil HCl & divide into 2 halves

(i) Add KCN (Red) (Blood)

(ii) Add $K_4[Fe(CN)_6]$ (Blue)

1 Reddish ppt form (KCN)

2 Prussian blue ($K_4[Fe(CN)_6]$)

Acetate ion
 CH_3COO^-

Salt + conc. $\text{H}_2\text{SO}_4 \rightarrow$ Colorless vapour w/
strong acetic smell

Confirmatory test

Salt + conc. H_2SO_4 + Ethanol $\rightarrow$ white ppt

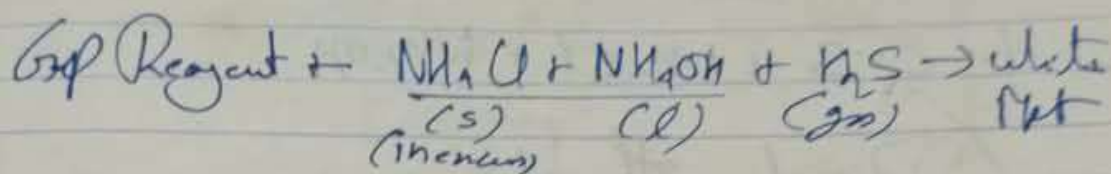
WE + few drops of neutral $\text{FeCl}_3 \rightarrow$ Deep
red color

Divide into 2 parts

i) Part I + dil $\text{HCl} \rightarrow$ Red precipitate

ii) Part II + add H_2O + boil $\rightarrow$ Reddish
Brown ppt

Group (IV) Zn^{2+}



Grp	Obs.	Inference
1) O.S + Grp reagent	white ppt	Zn^{2+}

$\Rightarrow$ dissolve the ppt in dil. HCl & divide into 2 halves.

i) Potassium Ferricyanide	Bleuish white ppt
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ii) NaOH drop wise	white ppt dissolves in sol ⁿ
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Zn^{2+} Confirmed